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SUPERSEDING
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MILITARY SPECIFICATION

PROJECTILE FOR CARTRIDGE, 81-MILLIMETER MORTAR, SMOKE, M370; WP FILLING ASSEMBLY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the filling of an 81-millimeter smoke projectile with white phosphorus (WP).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-46527 — Projectile, Smoke, WP, 81 MM, M370, Metal Parts Assembly.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

76-1-1734 — Marking Diagram for Pallet for Filled Projectile, 75 MM, 2120.

8846606 — Projectile, Smoke, WP, 81 MM, M370, Loading Assembly.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. All materials cited on drawing 8846606 and on the subsidiary drawings shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings.

3.2 Filling. The projectile, clean and free from foreign matter, shall be filled with molten WP. When water is in contact with

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the WP during the filling operation, the water used shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. The depth of water remaining on the surface of the WP in the filled and closed projectile shall not exceed 1/8 inch. The filling shall be solidified with the projectile in a nose-up position (see 6.2 and 6.3).

3.3 **Assembly.** Closing, coating, and marking of the projectile shall be in accordance with drawing 8846606, except that the burster and pad shall not be included in the assembly. The fuze and base threads shall be coated with silicone compound.

3.4 **Filling height.** When examined as specified in 4.6.1, the surface of the WP in the filled and closed projectile shall be 0 to 5/16 inch below the lowest point on the adapter. The projectile for cartridge shall be in a nose-up position and the WP at a temperature of 130° ± 10°F. (see 6.3).

3.5 **Cleaning after closing.** The exterior surface of the filled and closed projectile and the interior of the burster casing shall be thoroughly cleaned to remove any grease, oil, WP, or other contamination (see 6.4).

3.6 **Leakage.** The cleaned projectile shall not leak when tested as specified in 4.6.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the projectiles filled and assembled by one manufacturer, at one plant, using the same materials, under essentially the same filling conditions. However, no more than one lot of metal parts as identified by one manufacturer and one metal interfix number shall be represented in any one lot of filled projectiles.

4.4 **Sampling for nondestructive examination.** Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.5 **Examination.** Sample projectiles and the level B preparation for delivery thereof shall be examined in accordance with the classification of defects and Standard MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Projectile, smoke, WP, 81 mm, M370, loading assembly (Dwg 8846606).*

Categories Critical:	Defects	Acceptance standards
1	Leakage	4.6.2
2	WP contamination (3.5)	

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Major:		
	AQL 1.0 percent defective	
101	Burster casing not seated.	
102	Thread damaged.	
103	Marking (identification) incorrect.	
104	Bourrelet diameter (max) after coating incorrect.	
105	Excess water in projectile	4.6.1
106	Coating missing, inadequate, or incorrect	
107	Filling excessive	4.6.1

*Each projectile in each lot shall be examined for these defects.

Categories	*Defects*	*Acceptance standards*
Minor:	AQL 2.5 percent defective	
201	Marking (identification) missing or illegible.	
202	Burster casing contaminated.	
203	Fuze and base threads not coated.	
204	Closing plug frozen (when applicable).	
205	Closing plug missing (when applicable).	
206	Filling insufficient.	4.6.1

4.5.1.2 *Preparation for delivery (sec. 5).*

Categories	*Defects*
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Pallet incorrect.
102	Quantity per pallet incorrect.
103	Strapping loose or missing.
104	Damaged pallet (broken or wracked).
105	Marking incorrect, missing, or illegible.

4.6 Tests.

4.6.1 *Filling level.* The filling level shall be located radiographically with the projectile in a vertical, nose-up position.

4.6.2 *Leakage.* During or following the final cleaning operations and while the filling is still molten, the filled and closed projectile shall be placed in a nose-down position or rolled on its side for a period of no less than 1 minute. The projectile shall be observed for the emission of smoke which constitutes leakage.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level B.* One hundred rounds of filled and closed projectiles shall be prepared for shipment in accordance with the applicable requirements of Specification MIL-P-46527.

5.2 Marking. Marking shall be as shown on drawing 76-1-1734. The following additional marking shall be indicated on each pack of 100 rounds:

"100 PROJECTILES, 81 MM,
MORTAR, WP, M370, CU 17.1"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Supplier need not furnish radiographic equipment.

6.2 Precautions. Because of the hazardous nature of WP, filling installations are advised to take all possible precautions to protect personnel from injury while filling projectiles in accordance with this specification.

6.3 Filling aid. Projectiles filled to a height of 3⅜ to 3$^{9}/_{16}$ inches below the nose or with

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approximately 28 ounces of WP have been found to comply with 3.4, provided the projectiles meet the dimensional requirements of the applicable drawings. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 **Decontamination.** Projectiles may be decontaminated by flushing with hot water while scrubbing vigorously with a wire brush.

6.5 **Safety classification.** This item is chemical group C, explosive hazard class 2 (Chemical Corps Safety Directive No. 385-10).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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